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Transient response of rotating laminated functionally graded cylindrical shells in thermal environment

P. Malekzadeh [a,*], Y. Heydarpour [a], M.R. Golbahar Haghighi [a], M. Vaghefi [b]

[a] Department of Mechanical Engineering, School of Engineering, Persian Gulf University, Bushehr 75168, Iran
[b] Department of Civil Engineering, School of Engineering, Persian Gulf University, Bushehr 75168, Iran

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ABSTRACT

Based on the elasticity theory, the transient analysis of dynamically pressurized rotating multi-layered functionally graded (FG) cylindrical shells in thermal environment is presented. The variations of the field variables across the shell thickness are accurately modeled by dividing the shell into a set of co-axial mathematical layers in the radial direction. The initial thermo-mechanical stresses are obtained by solving the thermoelastic equilibrium equations. The differential quadrature method and Newmark's time integration scheme are employed to discretize the obtained governing equations of each mathematical layer. After performing the convergence and comparison studies, parametric studies for two common types of FG sandwich shells, namely, the shell with homogeneous inner/outer layers and FG core and the shell with FG inner/outer layers and homogeneous core are carried out. The influences of the temperature dependence of material properties, material graded index, the convective heat transfer coefficient, the angular velocity, the boundary condition and the geometrical parameters (length and thickness to outer radius ratios) on the dynamic response of the FG shells are investigated.

1. Introduction

Functionally graded materials (FGMs) are a new generation of composites which have found extensive applications in various industries due to their favorable and continuously varying physical and thermal properties [1–3]. Usually, these materials are made from a mixture of ceramic and metal. The ceramic constituent of the material provides the high temperature resistance due to its low thermal conductivity. The ductile metal constituent, on the order hand, prevents fracture caused by stresses due to high temperature gradient.

Currently, rotating functionally graded (FG) cylindrical shells are used in modern industries like aviation, rocketry, missiles, chemical, aero-space, mechanical and nuclear industries. Usually, these structures have been developed for using in high temperature environments. In addition, their working internal pressure may vary with time. Hence, for the purpose of their engineering design and manufacture, the transient dynamic analysis is essential for accurate evaluation of the induced stresses and deformations under these working conditions.

The static, buckling, free vibration and dynamic analysis of FG cylindrical shells under mechanical and thermal loading have been investigated by many researchers; see for example refs. [1–11]. In comparison to the research works in these topics, only limited studies can be found on the mechanical or thermo-mechanical analysis of FG rotating cylindrical shells [12–18].

Zenkour et al. [12] developed a one-dimensional analytical solution for the rotating FG hollow and solid long cylinders based on the plane strain assumption. In addition, a viscoelastic solution to the rotating viscoelastic cylinder was presented and dependence of stresses in hollow and solid cylinders on the time parameter was examined. Zamani Nejad and Rahimi [13] presented a one-dimensional analytical solution for the stresses in a rotating thick-walled FG cylindrical shell using elasticity theory subjected to plane stress and strain conditions. The pressure, thickness and Poisson's ratio were assumed to be constant but the other material properties vary in the radial direction according to power law. Khorshidvand and Khalili [14] obtained an analytical solution for the one-dimensional mechanical and thermal stresses in an FG rotating thick cylinder. Ghafoori and Asghari [15] proposed an elasticity solution for the analysis of FG rotating cylinders with variable thickness profile. The axisymmetric structure has been divided in several divisions in the radial direction. They assumed constant mechanical properties and

* Corresponding author. Tel.: +98 771 4222150; fax: +98 771 4540376.
E-mail addresses: p_malekz@yahoo.com, malekzadeh@pgu.ac.ir (P. Malekzadeh).

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thickness profile within each division to obtain an analytical solution. Akbarzadeh et al. [16] investigated the one-dimensional dynamic response of a rotating radially polarized FG piezoelectric hollow cylinder subjected to a thermo-electro-mechanical loading, which was placed in a constant magnetic field. All material properties were taken to follow a power law along the radial direction of the cylinder. Dai et al. [17] extracted an exact one-dimensional solution for a rotating FG piezoelectric hollow cylinder subjected to electric, thermal and mechanical loads. The problem was studied on the basis of the infinitesimal theory of electrothermoelasticity. More recently, Heydarpour et al. [18] investigated the steady state thermoelastic behavior of the rotating laminated FG cylindrical shells in thermal environment by employing an accurate and efficient solution procedure based on the elasticity theory. The material properties were assumed to be temperature-dependent and graded in the thickness direction. In this work, the steady state thermo-mechanical displacement and stress components due to rotation and thermal environment were evaluated.

From the above literature survey it is revealed that the transient deformation analysis of the rotating laminated FG cylindrical shells in thermal environment subjected to dynamic pressure is not investigated yet. Due to practical importance of this problem, it is studied in this paper. For this purpose, the layerwise approach is employed to divide the shell into some mathematical layers of arbitrary thickness and the related governing equations of each layer and method of solution are described in Section 2. Firstly, the heat transfer equation is solved in each individual layer to obtain the temperature distribution throughout the laminated FG cylinder in Section 2.1. Then, by solving the thermoelastic equilibrium equations and by considering the body forces due to rotation of the shell, the initial thermo-mechanical stresses are obtained. Considering these initial stresses and using Hamilton's principle in conjunction with the layerwise theory, the equations of motion for each individual layer are derived in Section 2.2. The differential quadrature method (DQM), as an efficient and accurate numerical tool [18–23], is utilized to discretize the governing thermoelastic equilibrium equations as well as equations of

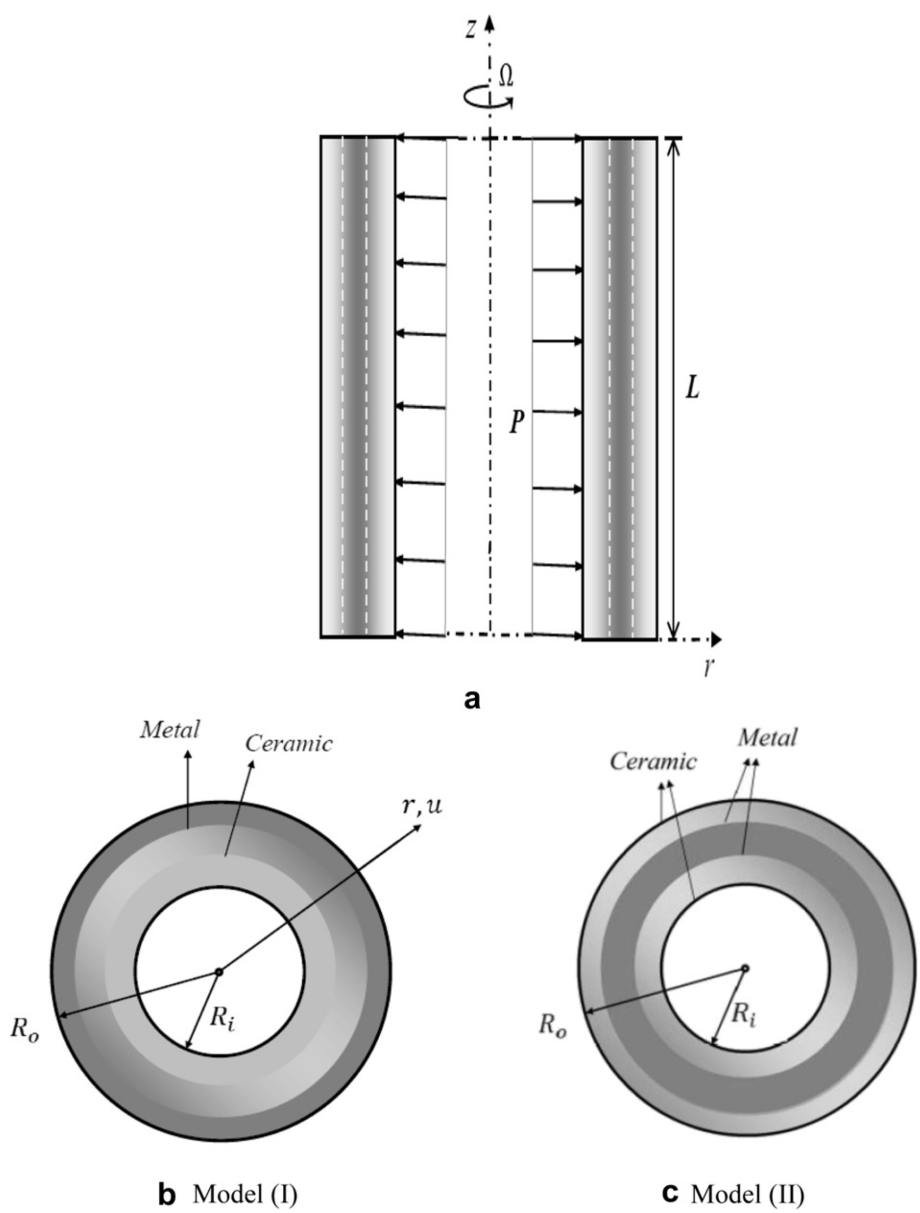

Fig. 1. Geometry of laminated FG cylindrical shells.

Table 1
Temperature-dependent coefficients of material properties for ceramic (ZrO_2) and metals (Ti–6Al–4V) [25].

	Material	Q_{-1}	Q_0	Q_1	Q_2	Q_3
E	Ti–6Al–4V	0	122.7 (GPa)	-4.605×10^{-4}	0	0
	ZrO_2	0	132.2 (GPa)	-3.805×10^{-4}	-6.127×10^{-8}	0
ν	Ti–6Al–4V	0	0.2888	1.108×10^{-4}	0	0
	ZrO_2	0	0.3330	0	0	0
ρ	Ti–6Al–4V	0	4420 (kg/m^3)	0	0	0
	ZrO_2	0	3657 (kg/m^3)	0	0	0
α	Ti–6Al–4V	0	7.43×10^{-6} (1/K)	7.483×10^{-4}	-3.621×10^{-7}	0
	ZrO_2	0	13.3×10^{-6} (1/K)	-1.421×10^{-3}	9.549×10^{-7}	0
κ	Ti–6Al–4V	0	6.10 (W/mK)	0	0	0
	ZrO_2	0	1.78 (W/mK)	0	0	0

Table 2
Convergence of the non-dimensional radial displacement of a single layer free ends rotating FG cylindrical shell ($h/R_o = 0.2$, $L/R_o = 1$, $R_o = 1$ m, $p = 1$, $\xi = \eta = 0.5$, $N_e = 1$).

N_z^e	t^*	$\Omega = 100$(Rad/s)			$\Omega = 400$(Rad/s)		
		$N_r^e = 11$	$N_r^e = 13$	$N_r^e = 15$	$N_r^e = 11$	$N_r^e = 13$	$N_r^e = 15$
7	0.1	−0.5694	−11.0950	3.9651	0.6912	−5.6643	3.4267
11		0.5515	0.5419	0.5464	1.3687	1.3627	1.3654
15		0.5494	0.5503	0.5470	1.3676	1.3680	1.3665
17		0.5494	0.5496	0.5495	1.3675	1.3676	1.3676
19		0.5494	0.5494	0.5495	1.3675	1.3676	1.3676
7	0.3	−0.2140	−10.7839	4.3169	0.9027	−5.4563	3.6349
11		0.9060	0.8968	0.9014	1.5790	1.5729	1.5757
15		0.9038	0.9049	0.9014	1.5779	1.5782	1.5766
17		0.9037	0.9040	0.9040	1.5777	1.5779	1.5779
19		0.9037	0.9039	0.9040	1.5777	1.5778	1.5779
7	0.5	−0.0781	−10.6660	4.4519	0.9849	−5.3746	3.7156
11		1.0414	1.0324	1.0371	1.6608	1.6546	1.6574
15		1.0393	1.0404	1.0369	1.6596	1.6599	1.6583
17		1.0392	1.0395	1.0394	1.6594	1.6596	1.6596
19		1.0392	1.0394	1.0394	1.6595	1.6595	1.6596

motion in the spatial domain. The resulting ordinary differential equations of motion are then solved in the temporal domain using Newmark's time integration scheme [24]. After demonstrating the correctness of the formulation together with the convergence and accuracy of the method of solution, two common types of FG sandwich cylindrical shells, namely, the sandwich cylindrical shell with homogeneous inner and outer layers and FG core and the sandwich cylinder shell with FG inner and outer layers and homogeneous core are analyzed in Section 3. The effect of angular velocity, thickness-to-outer radius and length ratios, layered layout, material properties and the convection heat transfer coefficient on the displacement and stress components and their time histories at different points of the FG cylinders are investigated.

Table 3
Convergence of the results for the rotating free ends FG cylindrical shell of model (II) ($L/R_o = 1, R_o = 1m, h/R_o = 0.2, N_z^e = 27, N_e = 3, \xi = \eta = 0.5p = 1$).

N_r^e	t^*	$\Omega = 100$(Rad/s)				$\Omega = 400$(Rad/s)			
		U	$\Sigma_{\theta\theta}$	Σ_{zz}	Σ_{rr}	U	$\Sigma_{\theta\theta}$	Σ_{zz}	Σ_{rr}
7	0.3	1.073	0.309	−0.068	−0.063	1.766	0.914	−0.046	−0.030
9		1.081	0.315	−0.066	−0.062	1.771	0.917	−0.045	−0.030
11		1.082	0.315	−0.067	−0.063	1.771	0.917	−0.045	−0.030
13		1.082	0.315	−0.067	−0.063	1.771	0.918	−0.045	−0.030
7	0.5	1.216	0.399	−0.068	−0.073	1.852	0.968	−0.046	−0.037
9		1.224	0.405	−0.066	−0.073	1.857	0.971	−0.046	−0.036
11		1.225	0.404	−0.068	−0.073	1.858	0.971	−0.046	−0.036
13		1.225	0.404	−0.068	−0.073	1.858	0.971	−0.046	−0.036

Table 4
Convergence of the results for the rotating free ends FG cylindrical shell of model (II) ($L/R_o = 1, R_o = 1m, h/R_o = 0.2, N_r^e = 13, N_e = 3, \xi = \eta = 0.5p = 1$).

N_z^e	t^*	$\Omega = 100$(Rad/s)				$\Omega = 400$(Rad/s)			
		U	$\Sigma_{\theta\theta}$	Σ_{zz}	Σ_{rr}	U	$\Sigma_{\theta\theta}$	Σ_{zz}	Σ_{rr}
15	0.3	1.081	0.312	−0.072	−0.066	1.771	0.919	−0.043	−0.029
19		1.082	0.315	−0.068	−0.063	1.770	0.917	−0.045	−0.030
21		1.082	0.315	−0.067	−0.063	1.770	0.917	−0.044	−0.032
25		1.082	0.315	−0.067	−0.063	1.771	0.918	−0.045	−0.030
27		1.082	0.315	−0.067	−0.063	1.771	0.918	−0.045	−0.030
15	0.5	1.223	0.400	−0.074	−0.077	1.858	0.973	−0.043	−0.035
19		1.225	0.404	−0.069	−0.074	1.857	0.972	−0.045	−0.036
21		1.225	0.404	−0.067	−0.073	1.857	0.972	−0.043	−0.036
25		1.225	0.404	−0.068	−0.073	1.858	0.971	−0.046	−0.036
27		1.225	0.404	−0.068	−0.073	1.858	0.971	−0.046	−0.036

2. Mathematical modeling

Consider a multi-layered FG cylindrical shell of the length L, inner and outer radius of R_i and R_o, respectively (Fig. 1). It is assumed that it is composed of N_L perfectly bonded physical layers and rotates about its axis with a constant angular velocity Ω (Fig. 1). Its inner surface is assumed to be subjected to an axisymmetric dynamic pressure and convection heat transfer but its outer surface is free from mechanical loading and has a constant temperature. The material properties and thickness of each layer of the shell are assumed to be arbitrary. Due to axisymmetric geometry and loading conditions of the shell, the thermoelastic equations reduce to the two-dimensional ones. Consequently, a cylindrical coordinate system (r,z) is sufficient to label the material points of the cylinder in the undeformed reference configuration.

In order to accurately model the variation of the field variables across the shell thickness, the shell is divided into $N_e(\geq N_L)$ co-axial mathematical layers in the radial direction with inner radius R_i^e and outer radius R_o^e. Hereafter, a superscript 'e' over a variable denotes the corresponding variable of the eth layer.

The material properties of each physical layer of the shell vary continuously and smoothly in the thickness direction (r) according to the power law distribution. Hence, a typical effective material property 'P' of the eth physical layer can be represented as,

$$P^e(r, T^e) = P_a^e(T^e) + \left[P_b^e(T^e) - P_a^e(T^e)\right]\left(\frac{r - R_i^e}{R_o^e - R_i^e}\right)^p \tag{1}$$

where $p(\geq 0)$ is the power law index (or the material property graded index) and T^e denotes the temperature (in Kelvin) at an arbitrary material point of the eth layer; also, if the inner surface of the eth layer is metal rich then $a = m$ and $b = c$ otherwise, if the inner surface is ceramic rich then $a = c$ and $b = m$, where the

Table 5
Comparison of the results for the freely rotating FG annular disk subjected to non-uniform temperature rise [$R_i/R_o = 0.2$, $L/R_o = 0.05$, $R_o = 1$ m, $\Omega = 600$(Rad/s), $\eta = 0.5$].

n	ξ	Present			Analytic [26]		
		U	$\Sigma_{\theta\theta}$	Σ_{rr}	U	$\Sigma_{\theta\theta}$	Σ_{rr}
0.5	0	0.1813	0.3691	1.41E-4	0.1813	0.3689	0
	0.5	0.2546	0.1730	0.1290	0.2547	0.1735	0.1294
	1	0.4054	0.0101	−1.24E-4	0.4054	0.0102	0
0	0	0.1470	0.6675	−1.51E-4	0.1468	0.6680	0
	0.5	0.2381	0.1639	0.1583	0.2381	0.1636	0.1580
	1	0.3997	0.005	−3.62E-4	0.3994	4.73E-3	0
−0.5	0	0.1197	1.2163	−4.60E-4	0.1194	1.2152	0
	0.5	0.2267	0.1464	0.1891	0.2267	0.1465	0.1893
	1	0.3965	0.0021	−1.43E-5	0.3961	1.74E-3	0

Table 6
Comparison of the results for the clamped ends FG cylindrical shell subjected to dynamic pressure [$R_o = 0.1$(m), $R_i = 0.08$(m), $L = 1$(m), $\xi = 0$, $\eta = 0.5$].

Source	t^*	U^*	$\Sigma^*_{\theta\theta}$	Σ^*_{zz}	Σ^*_{rr}
Present	0.1	0.202	1.102	0.231	−0.309
ANSYS [11]		0.201	1.109	0.232	−0.296
LWDQ [11]		0.196	1.082	0.237	−0.290
Present	0.3	0.530	2.885	0.596	−0.809
ANSYS [11]		0.525	2.903	0.606	−0.774
LWDQ [11]		0.505	2.783	0.611	−0.748
Present	0.5	0.655	3.566	0.735	−1.000
ANSYS [11]		0.652	3.589	0.749	−0.956
LWDQ [11]		0.637	3.510	0.771	−0.945

subscripts m and c refer to the metal and ceramic constituents, respectively.

The temperature dependence of a typical material property 'Q^e' of the eth layer can be represented as [25],

$$Q^e(T) = Q_0^e\left(Q_{-1}^e T^{-1} + 1 + Q_1^e T + Q_2^e T^2 + Q_3^e T^3\right) \tag{2}$$

The coefficients Q_i^e ($i = -1,0,1,2,3$) are unique to the constituent materials.

2.1. Evaluation of initial thermo-mechanical stresses

The difference between the thermal conditions of the operational and manufacturing environments can adjust the initial temperature distribution in the FG shell, which together with the mechanical constraints at the boundaries and non-homogeneity of the material properties can induce some thermal stresses in the FG shell. These initial thermal stresses together with the initial mechanical stresses due to rotation of the FG shell, affect the transient response of the FG shell under operational dynamic loading conditions. Hence, before performing the dynamic analysis of the FG shell, these initial thermo-mechanical stresses should be determined. For this purpose, the temperature distribution within the shell should be evaluated firstly. It is assumed that the shell is stress free at the temperature T_0 and the temperature rise is uniform or varies across its thickness and no heat generation source exists within it. Hence, the temperature distribution along the thickness direction can be obtained by solving the following steady state one-dimensional heat transfer equation in each layer of the cylinder,

$$\frac{d}{dr}\left(r\kappa^e(r,T^e)\frac{dT^e}{dr}\right) = 0 \tag{3}$$

where $\kappa^e(r,T^e)$ is the thermal conductivity of the eth layer. Among the different thermal boundary conditions that can be considered at the inner and the outer surfaces of the shell, without loss of generality, the following thermal boundary conditions are considered at the inner and outer surfaces of it, respectively,

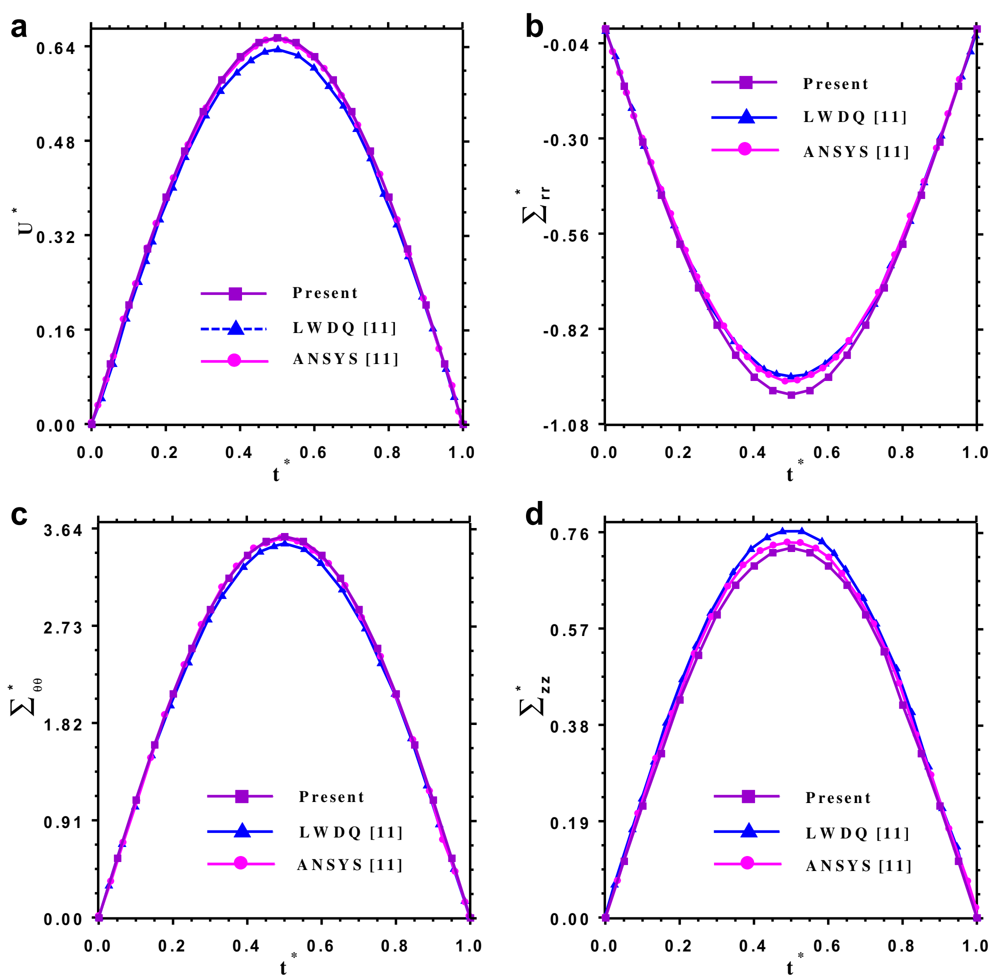

Fig. 2. Comparison of the stress and displacement components of the FG cylindrical shell subjected to dynamic pressure with those of ref. [11] [$R_o = 0.1$(m), $R_i = 0.08$(m), $L = 1$(m), $\xi = 0$, $\eta = 0.5$].

$$\text{At } r = R_i^e : -\kappa^e \frac{dT^e}{dr} = h_c(T_\infty - T^e) \quad \text{with } e = 1; \qquad \text{at } r = R_o^e : T^e = T_0 \quad \text{with } e = N_e \tag{4a, b}$$

where T_∞ and h_c are the inside fluid temperature and the convection heat transfer coefficient, respectively. In addition, at the interface of two adjacent layers one should satisfy the following conditions,

$$T(R_o^e) = T\left(R_i^{e+1}\right), \left[\kappa^e(r,T^e)\frac{dT^e}{dr}\right]_{r=R_o^e} = \left[\kappa^{e+1}\left(r,T^{e+1}\right)\frac{dT^{e+1}}{dr}\right]_{r=R_i^{e+1}} \quad \text{for } e=1,2,...,N_e-1 \tag{5a,b}$$

Using the DQM, Eq. (3) together with the boundary conditions (4) and compatibility conditions (5) can be discretized at the domain and boundary grid points of the layers, respectively [18]. After solving the resulting system of linear algebraic equations, the temperature distribution at each layer of the shell will be obtained.

Due to axisymmetric thermo-mechanical loading conditions, the thermoelastic equilibrium equations of each layer become axisymmetric. Under these conditions, the thermoelastic equilibrium equations based on the linear elasticity theory and in terms of the displacement components can be summarized as [18],

$$\begin{aligned} &C_{11}^e \frac{\partial^2 u_0^e}{\partial r^2} + \left(\frac{dC_{11}^e}{dr} + \frac{C_{11}^e}{r}\right)\frac{\partial u_0^e}{\partial r} + C_{55}^e \frac{\partial^2 u_0^e}{\partial z^2} + \left(\frac{dC_{12}^e}{dr} - \frac{C_{22}^e}{r} + \rho^e r \Omega^2\right)\frac{u_0^e}{r} \\ &+ (C_{55}^e + C_{13}^e)\frac{\partial^2 w_0^e}{\partial r \partial z} + \left(\frac{dC_{13}^e}{dr} + \frac{C_{13}^e - C_{23}^e}{r}\right)\frac{\partial w_0^e}{\partial z} \\ &= \left(\frac{dC_{11}^e}{dr} + \frac{dC_{12}^e}{dr} + \frac{dC_{13}^e}{dr}\right)\varepsilon_T^e + (C_{11}^e + C_{12}^e + C_{13}^e)\frac{d\varepsilon_T^e}{dr} - \rho^e r \Omega^2 \end{aligned} \tag{6}$$

$$\begin{aligned} &(C_{13}^e + C_{55}^e)\frac{\partial^2 u_0^e}{\partial r \partial z} + C_{55}^e \frac{\partial^2 w_0^e}{\partial r^2} + \left(\frac{dC_{55}^e}{dr} + \frac{C_{23}^e + C_{55}^e}{r}\right)\frac{\partial u_0^e}{\partial z} \\ &+ \left(\frac{dC_{55}^e}{dr} + \frac{C_{55}^e}{r}\right)\frac{\partial w_0^e}{\partial r} + C_{33}^e \frac{\partial^2 w_0^e}{\partial z^2} = 0 \end{aligned} \tag{7}$$

where u_0^e and w_0^e are the displacement components at an arbitrary material point of the eth layer in the thermoelastic equilibrium state along the r and z-directions, respectively; $\varepsilon_T^e(=\int_{T_0}^{T^e}\alpha^e dT)$ is the thermal strain and $\alpha^e[=\alpha^e(r,T^e)]$ is the thermal expansion coefficient; ρ^e is the mass density; $C_{ij}^e[=C_{ij}^e(r,T^e); i,j=1,2,3,5]$ are the

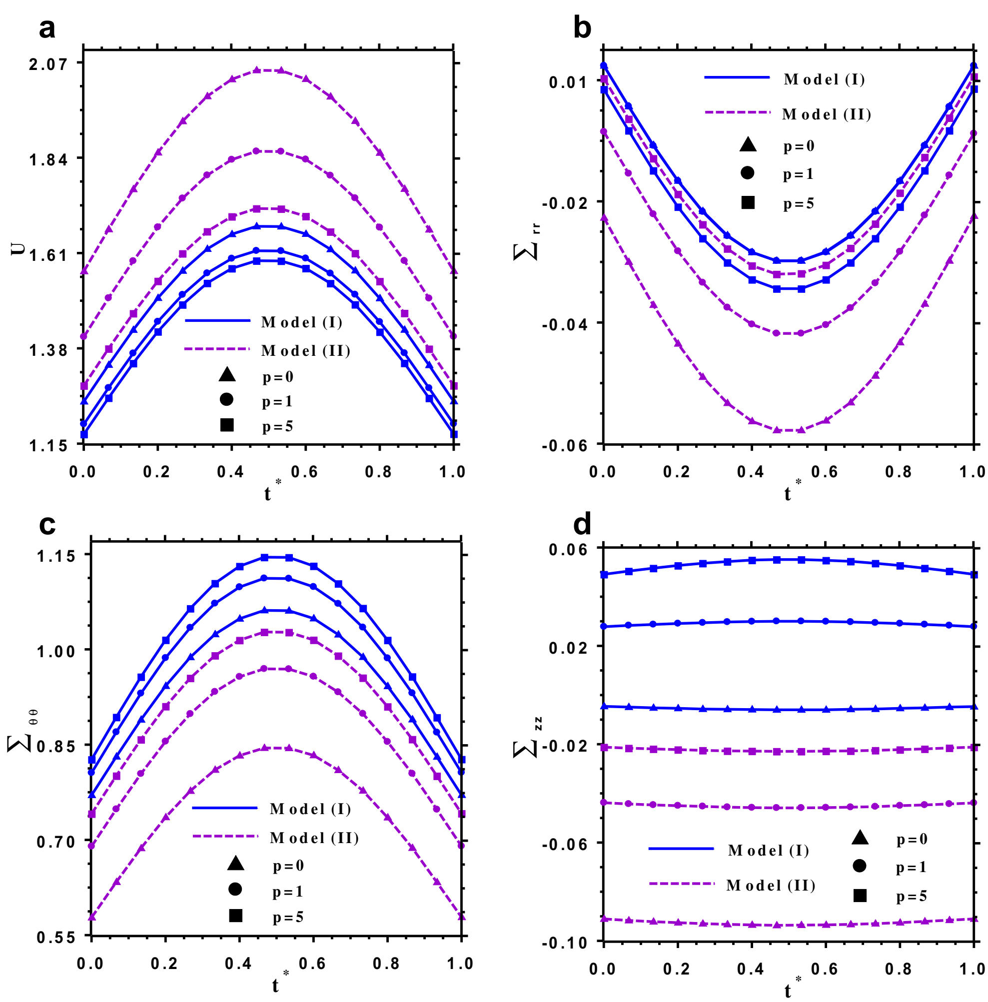

Fig. 3. The influence of the material graded index (p) on the time history of the results for the free ends FG cylindrical shells of model I and II [$h/R_o = 0.2, L/R_o = 1, R_o = 1m, N_e = 3, N_z^e = 27, N_r^e = 11, \Omega = 400(\text{Rad/s}), \xi = \eta = 0.5$].

material elastic coefficients, which for an isotropic FG material are related to Young's modulus $E^e(r,T^e)$ and Poisson's ratio $\nu^e(r,T^e)$ as,

$$C_{11}^e = C_{22}^e = C_{33}^e = \frac{(1-\nu^e)E^e}{(1+\nu^e)(1-2\nu^e)}, C_{12}^e = C_{23}^e = C_{13}^e = \frac{\nu^e E^e}{(1+\nu^e)(1-2\nu^e)}, C_{55}^e = \frac{E^e}{2(1+\nu^e)}, \tag{8a–c}$$

The external boundary conditions at the bottom and top surfaces$(z = 0,L)$ of the shell are

$$\text{Either } u_0^e = 0 \text{ or } C_{55}^e\left(\frac{\partial u_0^e}{\partial z}+\frac{\partial w_0^e}{\partial r}\right) = 0 \tag{9a, b}$$

$$\text{Either } w_0^e = 0 \text{ or } C_{13}^e\frac{\partial u_0^e}{\partial r}+\frac{C_{23}^e}{r}u_0^e+C_{33}^e\frac{\partial w_0^e}{\partial z}-\left(C_{13}^e+C_{23}^e+C_{33}^e\right)\varepsilon_T^e=0 \tag{10a,b}$$

Also, at the inner and outer surfaces of the shell $(r = R_i^1, R_o^{N_e})$ one has,

$$C_{11}^e\frac{\partial u_0^e}{\partial r}+\frac{C_{12}^e}{r}u_0^e+C_{13}^e\frac{\partial w_0^e}{\partial z}-\left(C_{11}^e+C_{12}^e+C_{13}^e\right)\varepsilon_T^e = 0, C_{55}^e\left(\frac{\partial u_0^e}{\partial z}+\frac{\partial w_0^e}{\partial r}\right) = 0 \tag{11a, b}$$

The thermoelastic geometrical and natural compatibility conditions at the interface of two adjacent layers of the shell are as follows,

$$u_0^e(R_o^e,z) = u_0^{e+1}\left(R_i^{e+1},z\right), w_0^e(R_o^e,z) = w_0^{e+1}\left(R_i^{e+1},z\right),$$
$$\left[C_{11}^e r\frac{\partial u_0^e}{\partial r}+C_{12}^e u_0^e+C_{13}^e r\frac{\partial w_0^e}{\partial z}-r\left(C_{11}^e+C_{12}^e+C_{13}^e\right)\varepsilon_T^e\right]_{r=R_o^e}$$
$$= \left[C_{11}^{e+1} r\frac{\partial u_0^{e+1}}{\partial r}+C_{12}^{e+1}u_0^{e+1}+C_{13}^{e+1}r\frac{\partial w_0^{e+1}}{\partial z} -r\left(C_{11}^{e+1}+C_{12}^{e+1}+C_{13}^{e+1}\right)\varepsilon_T^{e+1}\right]_{r=R_i^{e+1}}, C_{55}^e\left(\frac{\partial u_0^e}{\partial z}+\frac{\partial w_0^e}{\partial r}\right)\Bigg|_{r=R_o^e}$$
$$= C_{55}^{e+1}\left(\frac{\partial u_0^{e+1}}{\partial z}+\frac{\partial w_0^{e+1}}{\partial r}\right)\Bigg|_{r=R_i^{e+1}} \quad \text{for} \quad e = 1,2,\ldots,N_e-1 \tag{12a-d}$$

Again, using the DQM, the above mentioned differential equations and the related boundary and compatibility conditions are transformed into a set of algebraic equations [18]. After solving

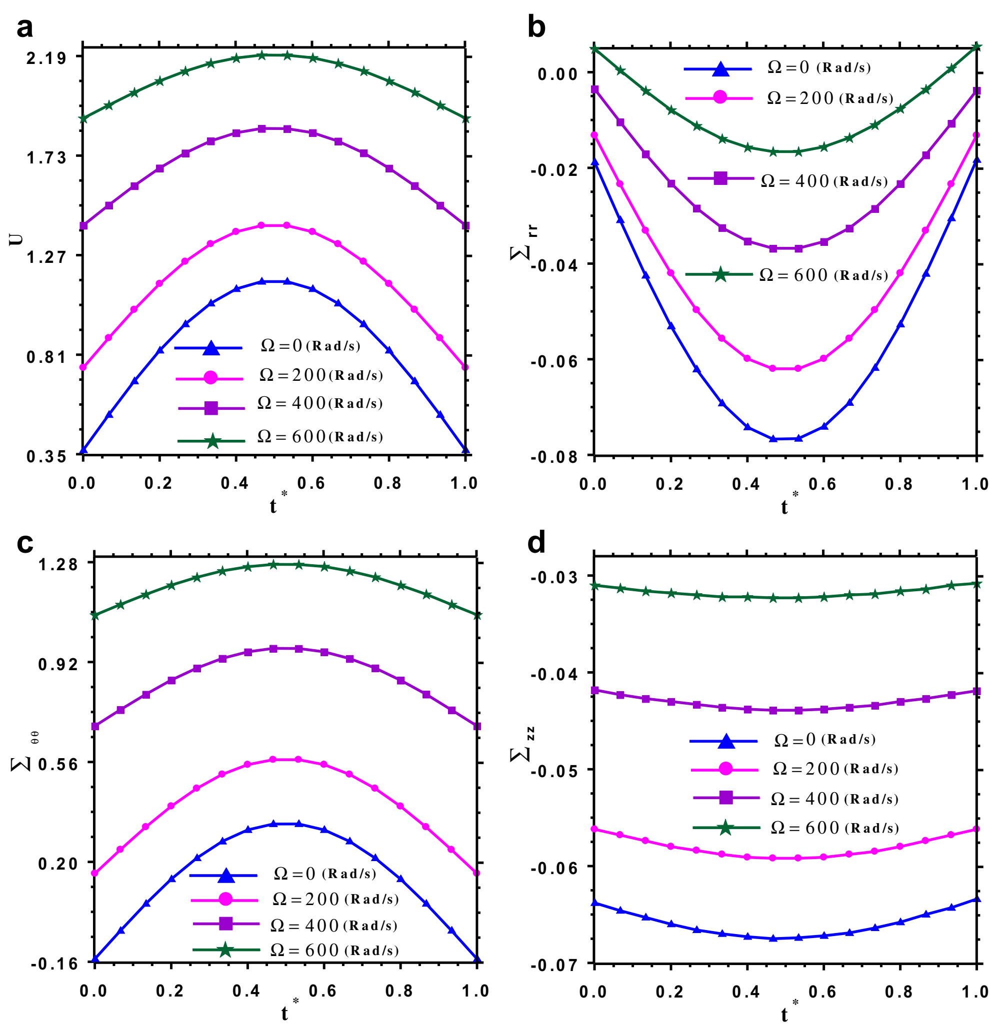

Fig. 4. The effect of angular velocity on the time history of the results for the free ends FG cylindrical shells of model (II) $[h/R_o = 0.2, L/R_o = 1, R_o = 1m, N_e = 3, N_z^e = 27, N_r^e = 11, p = 1, \xi = \eta = 0.5]$.

them, the displacement components and consequently the initial thermo-mechanical stresses will be obtained.

2.2. Transient response of the laminated FG cylindrical shells

If the shell, which steady stately rotates with a constant angular velocity in thermal environment, is subjected to an axisymmetric dynamic internal pressure, it undergoes some additional deformations. The displacement components due to this loading is measured from the thermoelastic equilibrium state and are denoted as $u^e(r,z,t)$ and $w^e(r,z,t)$ along the r and z-directions, respectively. Hence, the total displacement components along these directions measured from the shell un-deformed configuration become, respectively,

$$u_T^e(r,z,t) = u_0^e(r,z) + u^e(r,z,t), w_T^e(r,z,t) = w_0^e(r,z) + w^e(r,z,t) \quad (13a, b)$$

The equations of motion around the thermoelastic equilibrium state together with the related boundary and compatibility conditions at the interface of two adjacent layers of the shell can be obtained using Hamilton's principle,

$$\int_{t_1}^{t_2} (\delta K - \delta U + \delta W)dt = 0 \quad (14)$$

In the above equation, t_1 and t_2 are two arbitrary times; δ is the variational operator; K and U are the kinetic and potential energies of the FG. The variation of the potential and kinetic energy of the FG shell can be written as,

$$\delta U = 2\pi \int_0^L \int_{R_i}^{R_o} \Big[(\sigma_{rr}^e + \sigma_{0rr}^e)(\delta\varepsilon_{rr}^e + \delta\varepsilon_{0rr}^e) + (\sigma_{\theta\theta}^e + \sigma_{0\theta\theta}^e)(\delta\varepsilon_{\theta\theta}^e + \delta\varepsilon_{0\theta\theta}^e) + 2(\sigma_{rz}^e + \sigma_{0rz}^e)(\delta\varepsilon_{rz}^e + \delta\varepsilon_{0rz}^e)\Big] r dr dz \quad (15)$$

$$\delta K = 2\pi \int_0^L \int_{R_i}^{R_o} \rho^e \left(\frac{\partial u^e}{\partial t}\frac{\partial \delta u^e}{\partial t} + \frac{\partial w^e}{\partial t}\frac{\partial \delta w^e}{\partial t} + \Omega^2 u^e \delta u^e \right) r dr dz \quad (16)$$

where ε_{0ij}^e andσ_{0ij}^e $(i,j = r,\theta,z)$ are the initial strain and stress tensor components, respectively; ε_{ij}^e and σ_{ij}^e $(i,j = r,\theta,z)$ are also the strain and stress tensor components due to external transient loading, respectively. One can note that since the displacement components at the thermoelastic equilibrium state are known, then $\delta\varepsilon_{0ij}^e = 0$.

The virtual work (δW) of the internal dynamic pressure $P(z,t)$ is as follows,

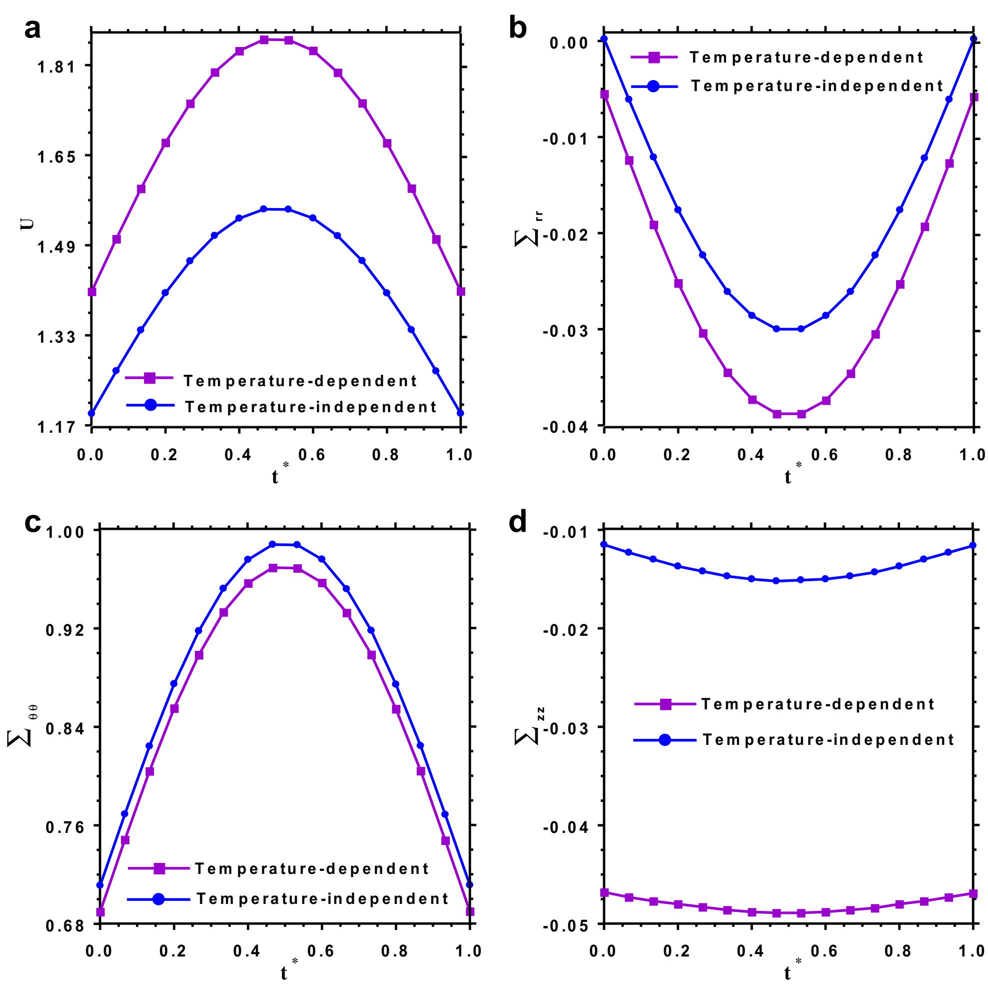

Fig. 5. The influence of the temperature dependence of material properties on the time history of the results for the free ends rotating FG cylindrical shells of model (II) $[h/R_o = 0.2, L/R_o = 1, R_o = 1m, N_e = 3, N_z^e = 27, N_r^e = 11, \Omega = 400(\mathrm{Rad/s}), p = 1, \xi = \eta = 0.5]$.

$$\delta W = 2\pi \int\limits_0^L P(z,t)\delta u^e\left(R_i^e, z, t\right)dz, e = 1 \tag{17}$$

Using the mechanical constitutive relations together with the thermoelastic equilibrium equations (6) and (7) and after performing the related integral by parts, the equations of motion and the related external boundary conditions and also the compatibility conditions at the interface of two adjacent layers of the shell can be obtained from Eq. (14) as follows,

Equations of motion:

$$\begin{aligned}\delta u^e :& \left(\frac{\partial C_{12}^e}{\partial r} - \frac{C_{22}^e}{r} - \frac{\sigma_{0\theta\theta}^e}{r} + r\rho^e\Omega^2\right) u^e + \left[C_{11}^e + r\frac{\partial C_{11}^e}{\partial r} + \sigma_{0\theta\theta}^e \right.\\ &\left. - \rho^e r\left(r + u_0^e\right)\Omega^2\right] \frac{\partial u^e}{\partial r} + \left(rC_{11}^e + r\sigma_{0rr}^e\right) \frac{\partial^2 u^e}{\partial r^2}\\ &+ \left(r\sigma_{0zz}^e + rC_{55}^e\right) \frac{\partial^2 u^e}{\partial z^2} + \left(C_{13}^e + r\frac{\partial C_{13}^e}{\partial r} - C_{23}^e\right) \frac{\partial w^e}{\partial z}\\ &+ 2r\sigma_{0rz}^e \frac{\partial^2 u^e}{\partial r \partial z} + \left(rC_{55}^e + rC_{13}^e\right) \frac{\partial^2 w^e}{\partial r \partial z} = r\rho^e \frac{\partial^2 u^e}{\partial t^2}\end{aligned} \tag{18}$$

$$\begin{aligned}\delta w^e :& \left[C_{55}^e + r\frac{\partial C_{55}^e}{\partial r} + \sigma_{0\theta\theta}^e - \rho^e r\left(r + u_0^e\right)\Omega^2\right] \frac{\partial w^e}{\partial r}\\ &+ \left(r\sigma_{0rr}^e + rC_{55}^e\right) \frac{\partial^2 w^e}{\partial r^2} + \left(rC_{33}^e + r\sigma_{ozz}^e\right) \frac{\partial^2 w^e}{\partial z^2}\\ &+ \left(C_{23}^e + C_{55}^e + r\frac{\partial C_{55}^e}{\partial r}\right) \frac{\partial u^e}{\partial z} + 2\left(r\sigma_{orz}^e\right) \frac{\partial^2 w^e}{\partial r \partial z}\\ &+ \left(rC_{13}^e + rC_{55}^e\right) \frac{\partial^2 u^e}{\partial r \partial z} = r\rho^e \frac{\partial^2 w^e}{\partial t^2}\end{aligned} \tag{19}$$

Boundary conditions on the surfaces $r = R_i^1 = R_i$ and $r = R_o^{N_e} = R_o$:

$$\begin{aligned}& rC_{11}^e \frac{\partial u^e}{\partial r} + C_{12}^e u^e + rC_{13}^e \frac{\partial w^e}{\partial z} + r\sigma_{0rr}^e \frac{\partial u^e}{\partial r} + r\sigma_{0rz}^e \frac{\partial u^e}{\partial z} = -P(z,t),\\ & r\sigma_{0rr}^e \frac{\partial w^e}{\partial r} + rC_{55}^e \left(\frac{\partial w^e}{\partial r} + \frac{\partial u^e}{\partial z}\right) + r\sigma_{0rz}^e \frac{\partial w^e}{\partial z} = 0\end{aligned} \tag{20a, b}$$

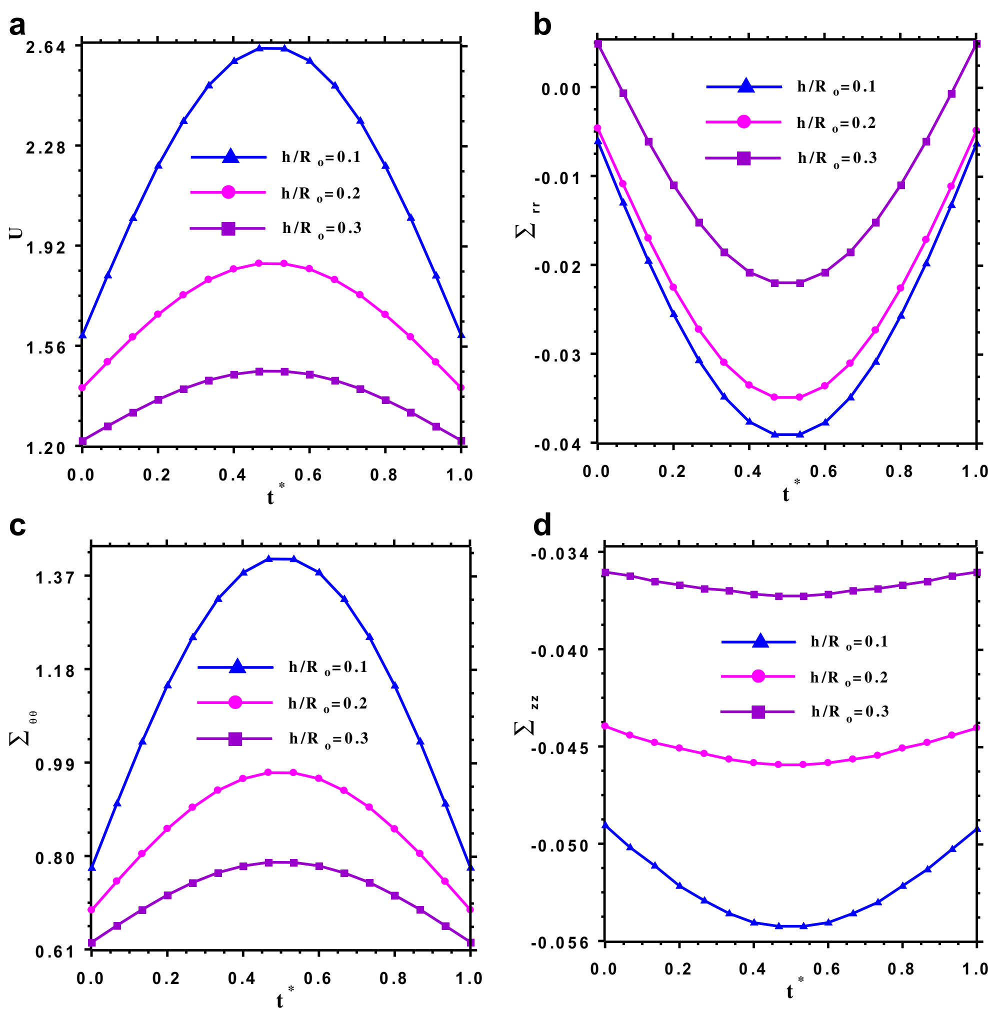

Fig. 6. The effect of thickness to outer radius ratio on the time history of the results for the free ends rotating FG cylindrical shells of model (II)[$h/R_o = 0.2, L/R_o = 1, R_o = 1m$, $N_e = 3, N_z^e = 27, N_r^e = 11, \Omega = 400(\text{Rad/s}), p = 1, \xi = \eta = 0.5$].

Geometrical and natural compatibility conditions at the interface of two adjacent layers:

$$u^e(R_o^e,z) = u^{e+1}\left(R_i^{e+1},z\right) w^e(R_o^e,z) = w^{e+1}\left(R_i^{e+1},z\right),$$
$$\left[rC_{11}^e\frac{\partial u^e}{\partial r}+C_{12}^e u^e+rC_{13}^e\frac{\partial w^e}{\partial z}+r\sigma_{0rr}^e\frac{\partial u^e}{\partial r}+r\sigma_{0rz}^e\frac{\partial u^e}{\partial z}\right]_{r=R_o^e}$$
$$=\left[rC_{11}^{e+1}\frac{\partial u^{e+1}}{\partial r}+C_{12}^{e+1}u^{e+1}+rC_{13}^{e+1}\frac{\partial w^{e+1}}{\partial z}+r\sigma_{0rr}^{e+1}\frac{\partial u^{e+1}}{\partial r}\right.$$
$$\left.+r\sigma_{0rz}^{e+1}\frac{\partial u^{e+1}}{\partial z}\right]_{r=R_o^{e+1}} \left[r\sigma_{0rr}^e\frac{\partial w^e}{\partial r}+rC_{55}^e\left(\frac{\partial w^e}{\partial r}+\frac{\partial u^e}{\partial z}\right)\right.$$
$$\left.+r\sigma_{0rz}^e\frac{\partial w^e}{\partial z}\right]_{r=R_o^e} = \left[r\sigma_{0rr}^{e+1}\frac{\partial w^{e+1}}{\partial r}+rC_{55}^{e+1}\left(\frac{\partial w^{e+1}}{\partial r}+\frac{\partial u^{e+1}}{\partial z}\right)\right.$$
$$\left.+r\sigma_{0rz}^{e+1}\frac{\partial w^{e+1}}{\partial z}\right]_{r=R_i^{e+1}} \tag{21a–d}$$

Without loss of the generality of formulation and method of solution, zero initial conditions are assumed,

$$u^e(r,z,0)=0, w^e(r,z,0)=0, \left.\frac{\partial u^e}{\partial t}\right|_{t=0}=0, \left.\frac{\partial w^e}{\partial t}\right|_{t=0}=0 \tag{22a–d}$$

Different types of the classical boundary conditions can be considered at the lower and upper surfaces of the shell (z = 0 and z = L),

$$\text{Simply support (S)}: u^e=0, rC_{13}^e\frac{\partial u^e}{\partial r}+C_{23}^e u^e+rC_{33}^e\frac{\partial w^e}{\partial z}+r\sigma_{0zz}^e\frac{\partial w^e}{\partial z}+r\sigma_{0rz}^e\frac{\partial w^e}{\partial r}=0 \tag{23a, b}$$

$$\text{Clamped (C)}: u^e = 0, w^e = 0 \tag{24a, b}$$

$$\text{Free (F)}: C_{55}^e\left(\frac{\partial w^e}{\partial r}+\frac{\partial u^e}{\partial z}\right)+\sigma_{0zz}^e\frac{\partial u^e}{\partial z}+\sigma_{0rz}^e\frac{\partial u^e}{\partial r}=0,$$
$$rC_{13}^e\frac{\partial u^e}{\partial r}+C_{23}^e u^e+rC_{33}^e\frac{\partial w^e}{\partial z}+r\sigma_{0zz}^e\frac{\partial w^e}{\partial z}+r\sigma_{0rz}^e\frac{\partial w^e}{\partial r}=0 \tag{25a, b}$$

where $e = 1,2,...,N_e$

It is obvious that the above systems of differential equations have variable coefficients and consequently if it is not impossible to solve them analytically, it is very difficult to obtain such a solution. Therefore, a couple of the differential quadrature method (DQM), which has been established as a simple, efficient and accurate numerical technique for different structural problems, especially, for those with variable coefficients differential equations [18–23], and Newmark's time integration scheme [24] is employed to discretize the spatial and temporal domain, respectively.

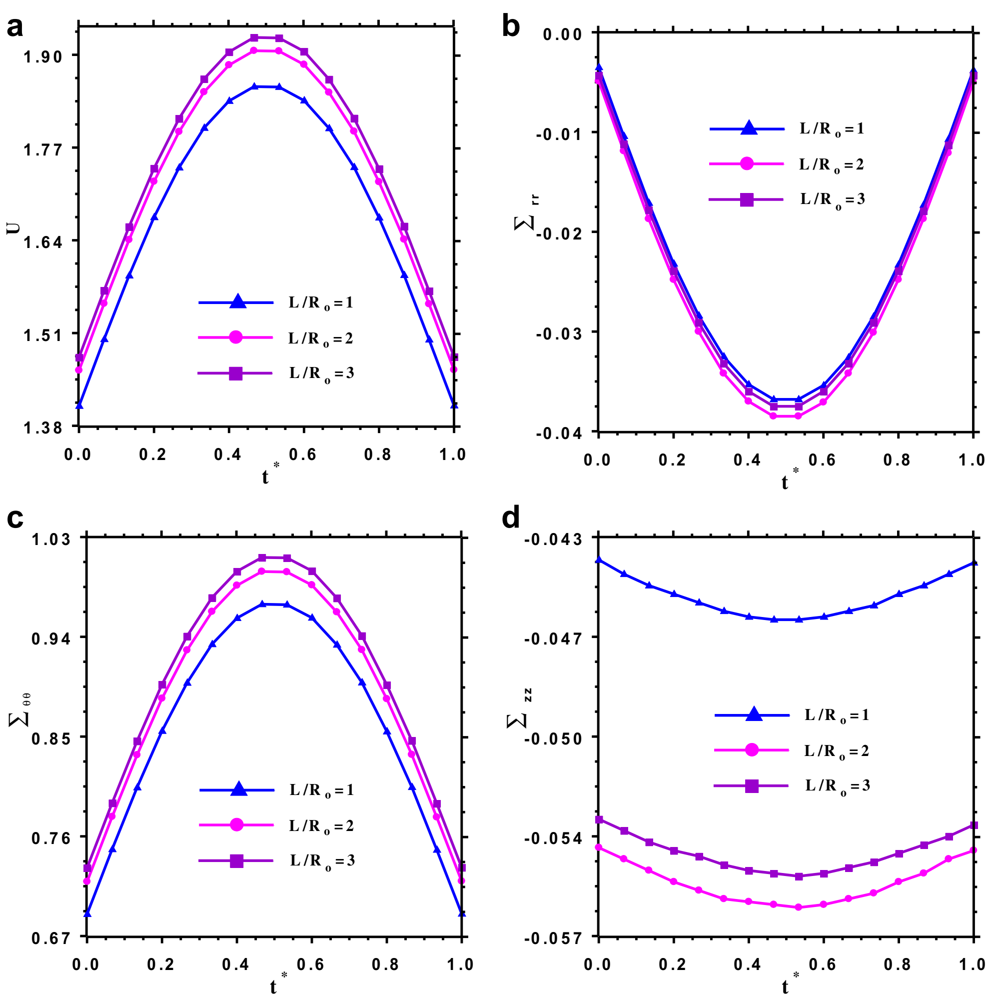

Fig. 7. The effect of length-to-outer radius ratio on the time history of the results for the free ends rotating FG cylindrical shells of model (II) [$h/R_o = 0.2, L/R_o = 1, R_o = 1m, N_e = 3, N_z^e = 27, N_r^e = 11, \Omega = 400(\text{Rad/s}), p = 1, \xi = \eta = 0.5$].

Based on the DQM, in the computational domain of each mathematical layer is discretized into N_r^e and N_z^e discrete grid points along the *r*- and *z*-directions, respectively. The cosine type grid generation rule is used in the both *r*- and *z*-directions [18–23]. Using the DQ discretization rules, at each domain grid points, the strong form of the equations of motion and at the boundary grid points, the boundary and at the interface of two adjacent layers, the compatibility conditions are discretized [18–23]. After doing these, the discretized governing differential equations and the associated boundary and compatibility conditions can be represented in the matrix form as,

$$\left[\widehat{M}\right]\left\{\frac{d^2D}{dt^2}\right\} + \left[\widehat{K}\right]\{D\} = \left\{\widehat{F}(t)\right\} \tag{26}$$

where $[\widehat{M}]$, $[\widehat{K}]$ and $\{\widehat{F}\}$ are the mass matrix, the stiffness matrix and the load vector, respectively; also {D} is the degrees of freedom vector, which contains the DQ grid points displacement components. Based on the element arrangement of this vector, the elements of the mass matrix, stiffness matrix and the load vector are obtained from the discretized form of the governing differential equations and the related boundary and compatibility conditions.

Employing the constant–average acceleration method form Newmark's time integration scheme family [24], the system of differential Eq. (26) subjected to the related initial conditions is solved in each time step. Consequently, the strain and stress components at any point of the FG cylindrical shell can be determined.

3. Numerical results

In this section, firstly, the convergence behavior of the method for the transient analysis of dynamically pressurized rotating FG laminated cylindrical shells in thermal environment is investigated. In order to verify the accuracy of the model, comparison studies with the other available solutions are performed. Then, parametric studies for two common types of FG sandwich shells, namely, the shell with homogeneous inner/outer layers and FG core [model (I), see Fig. 1(a)] and the shell with FG inner/outer layers and homogeneous core [model (II), see Fig. 1(b)] are carried out. In all cases, without loss of generality, mathematical layers with equal thickness and also equal number of grid points along the axial and radial directions are used in the numerical computations.

Otherwise specified, a dynamic pressure $P(z,t) = P_0 sin(\pi t/t_o)$, which is adopted from ref. [11], is assumed to be applied on the inner surface of the shell; in addition, the values of $P_0 = 100$(MPa), $t_o = 1$(s), $T_0 = 300$(K), $T_\infty = 1100$(K) and $h_c = 200$(W/m^{2}K) are used in the case studies. In order to perform the dynamic analysis under this pressure loading, a convergence study is performed. It is found that a time step of $\Delta t = 5 \times 10^{-3} t_0$ is appropriate to obtain the

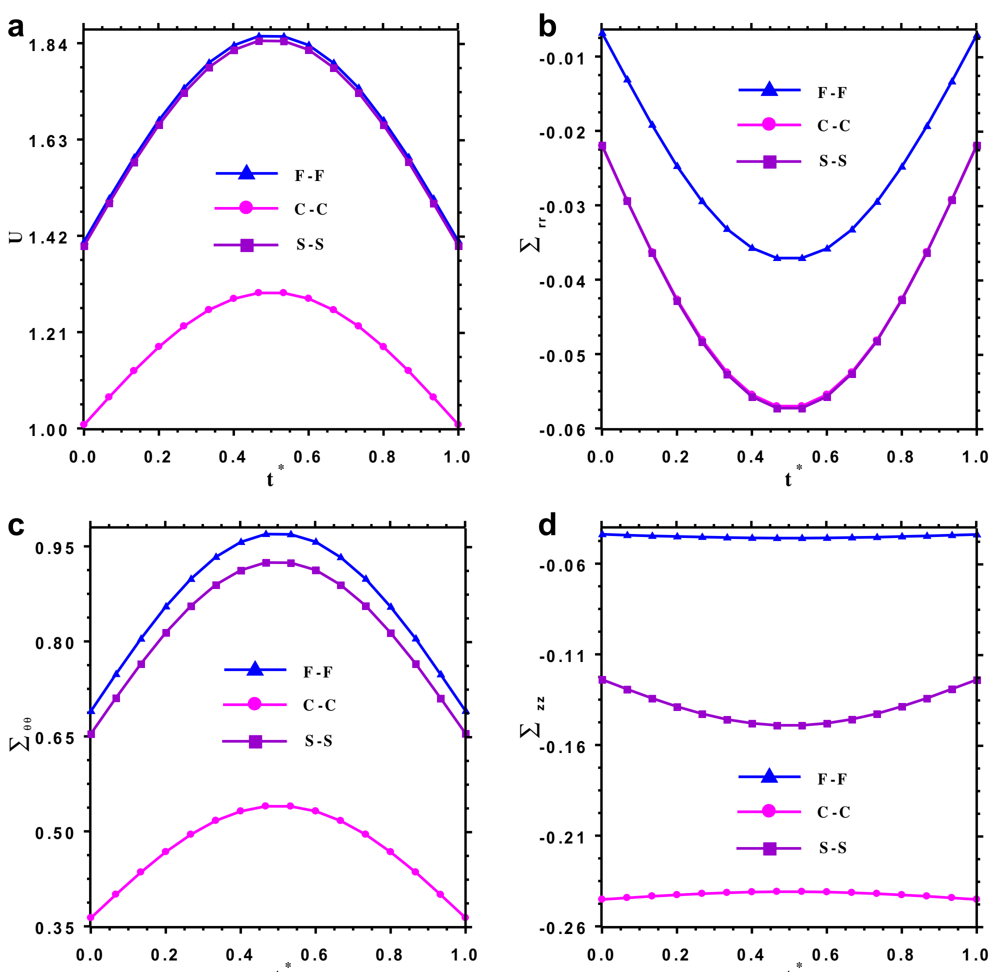

Fig. 8. The influence of different boundary conditions on the time history of the results for the rotating FG cylindrical shells of model (II) $[h/R_o = 0.2, L/R_o = 1, R_o = 1m, N_e = 3, N_z^e = 27, N_r^e = 11, \Omega = 400(\text{Rad/s}), p = 1, \xi = \eta = 0.5]$.

accurate converged results when using the constant-average acceleration method of Newmark's time integration scheme.

Except in the comparison studies, the temperature-dependent material properties are used and the non-dimensional parameters are defined as [11,26],

$$\xi = \frac{r-R_i}{R_o-R_i}, \eta = z/L, t^* = \frac{t}{t_o},$$
$$U = \frac{uE_{oc}}{(1+\nu_{oc})\left[E_{oc}\alpha_{oc}T_0+(1-\nu_{oc})\rho_c\Omega^2R_o^2\right]R_o}, \tag{27}$$
$$\Sigma_{ii} = \frac{(1-\nu_{oc})\sigma_{ii}}{E_{oc}\alpha_{oc}T_0+(1-\nu_{oc})\rho_c\Omega^2R_o^2} \text{ with } i = r,\theta,z$$

The material properties of Ti–6Al–4V and ZrO_2, as given in Table 1, are used to obtain the new numerical results. These material properties are chosen from the work of Kim [25] and are valid for the temperature range of 300 K $\le T \le$ 1100 K

As a first example, the convergence behavior of the method against the DQ number of grid points along the r- and z-direction is shown in Table 2 for predicting the non-dimensional radial displacement component of a single layer rotating FG cylindrical shell subjected to the previously specified dynamic pressure. The results are prepared at different non-dimensional times and for two different values of the angular velocity. It is obvious that in all cases, a single element with $N_r^e = 15$ and $N_z^e = 17$ is sufficient to obtain the converged results with at least four significant digits.

In Tables 3 and 4, the convergence behavior of the method for the rotating sandwich cylindrical shell of model (II) against the DQ number of grid points along the radial and axial directions is presented. Three mathematical layers are used to model the shell in the radial direction. In all cases, the fast rate of convergence of the method is quite evident and it is found that $N_r^e = 13$ and $N_z^e = 19$ is sufficient to obtain the accurate results.

In order to verify the presented approach for the rotating FG cylindrical shells in thermal environment, the available results for annular disk as a limiting case of a cylinder (cylinder with small length-to-outer radius ratio) is used. For this purpose, the numerical results for the rotating short FG cylinder ($L/R_o = 0.05$) subjected to thermal environment are compared with the analytical solution of Peng and Li [26]. They transformed the one-dimensional thermoelasticity equation into a Fredholm integral equation to obtain the analytical solution. To find such a solution, all the material properties, except Poisson's ratio, are assumed to vary according to $\psi = \psi_0 r^n$ where ψ_0 is a material constant at the outer surface and n is the material graded index. The material properties are as follows [26],

$E_m = 70$ GPa, $\nu_m = 0.3$, $K_m = 209$(W/m °C), $\alpha_m = 23 \times 10^{-6}$(1/°C), $\rho_m = 2700$(Kg/m^3),

$E_c = 151$ GPa, $\nu_c = 0.3$, $K_c = 2$(W/m °C), $\alpha_c = 10 \times 10^{-6}$(1/°C), $\rho_c = 5700$(Kg/m^3),

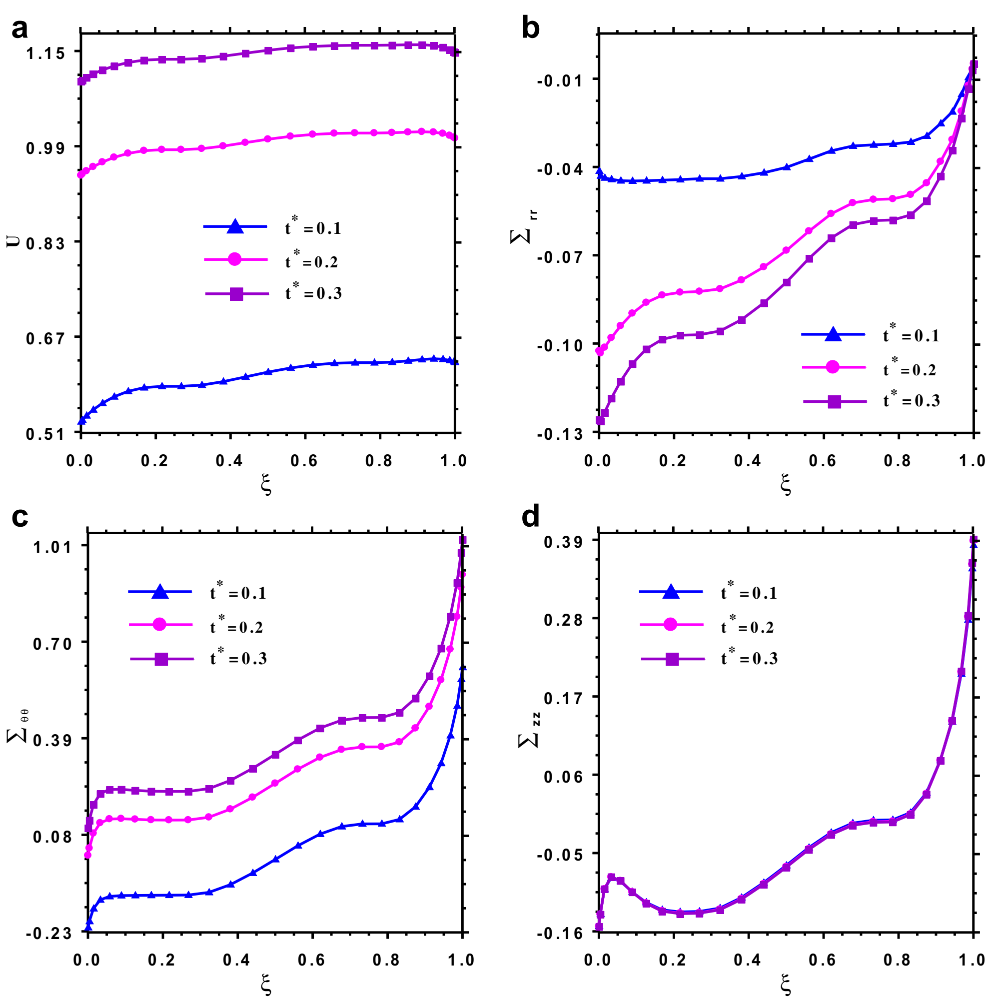

Fig. 9. Through the thickness variation of the results for the free ends FG cylindrical shells of model (II) at different time levels [$h/R_o = 0.2, L/R_o = 1, R_o = 1m, N_e = 3, N_z^e = 27, N_r^e = 11, \Omega = 400$(Rad/s), $p = 1, \xi = \eta = 0.5$].

Also the surface temperature at the inner and outer surfaces of disk is assumed to be.

$T(R_i) = 0\ °\mathrm{C}$, $T(R_o) = 1000\ °\mathrm{C}$

The non-dimensional radial displacement and stress components at different locations and for different values of the material graded index (n) are compared with those of Peng and Li [26] in Table 5. Excellent agreement of the results validates the presented approach.

In order to better clarify the accuracy of the presented approach, the transient analysis of a stationary FG cylindrical shell subjected to internal dynamic pressure, which has been analyzed by Setoodeh et al. [11], is examined here. They [11] obtained the dynamic response of an FG hollow cylinder under a uniformly distributed but transient pressure $P(t) = P_0 sin(\pi t)$ on its inner surface. Stress free conditions are assumed on the outer surface. They employed the layerwise finite element through the thickness of the shell and DQM along its axial direction (LWDQ). In addition, they modeled the problem using the commercial finite element software ANSYS. One-quadrant of the shell structure is simulated using 18000 3D solid elements with 20 nodes. They assumed that the modulus of elasticity and the mass density vary through the shell thickness according to the following relations, respectively,

$$E(r) = E_m\left(\frac{r}{R_i}\right)^{n_1}, \rho(r) = \rho_m\left(\frac{r}{R_i}\right)^{n_2} \tag{28a, b}$$

where E_m and ρ_m are the metal modulus of elasticity and density, respectively. Also, n_1 andn_2 are the material graded index. The material properties and also the non-dimensional parameters of the FG shell are as follows [11],

$$\begin{aligned} &E_m = 223\mathrm{GPa}, \rho_m = 8900\left(\mathrm{Kg/m^3}\right), \nu = 0.3, \\ &n_1 = 2, n_2 = -5.93, U^* = \frac{u}{P_0 h/k^*}, \Sigma^*_{ii} = \frac{\sigma_{ii}}{P_0}(i = r, \theta, z) \end{aligned} \tag{29}$$

where $k^* = 10$ GPa.

The numerical results for the clamped FG cylindrical shell under the above mentioned conditions are compared with the numerical solution of Setoodeh et al. [11] in Table 6 and Fig. 2. It is obvious that close agreement exists between the results of three approaches. However, the agreement between the present approach and ANSYS is better than those of LWDQ of ref. [11] and ANSYS. In addition, it is interesting to note that both ANSYS and LWDQ cannot predict the actual value of the radial stress on the inner surface of the shell, which should be equal to the applied pressure on this surface. However, the present approach gives its exact value. This is because in both ANSYS and LWDQ the weak form of the governing equations in the radial direction are discretized and consequently, the force boundary conditions are implemented through the weak form of the governing equations. However, in the present approach, the strong forms of the boundary conditions are discretized.

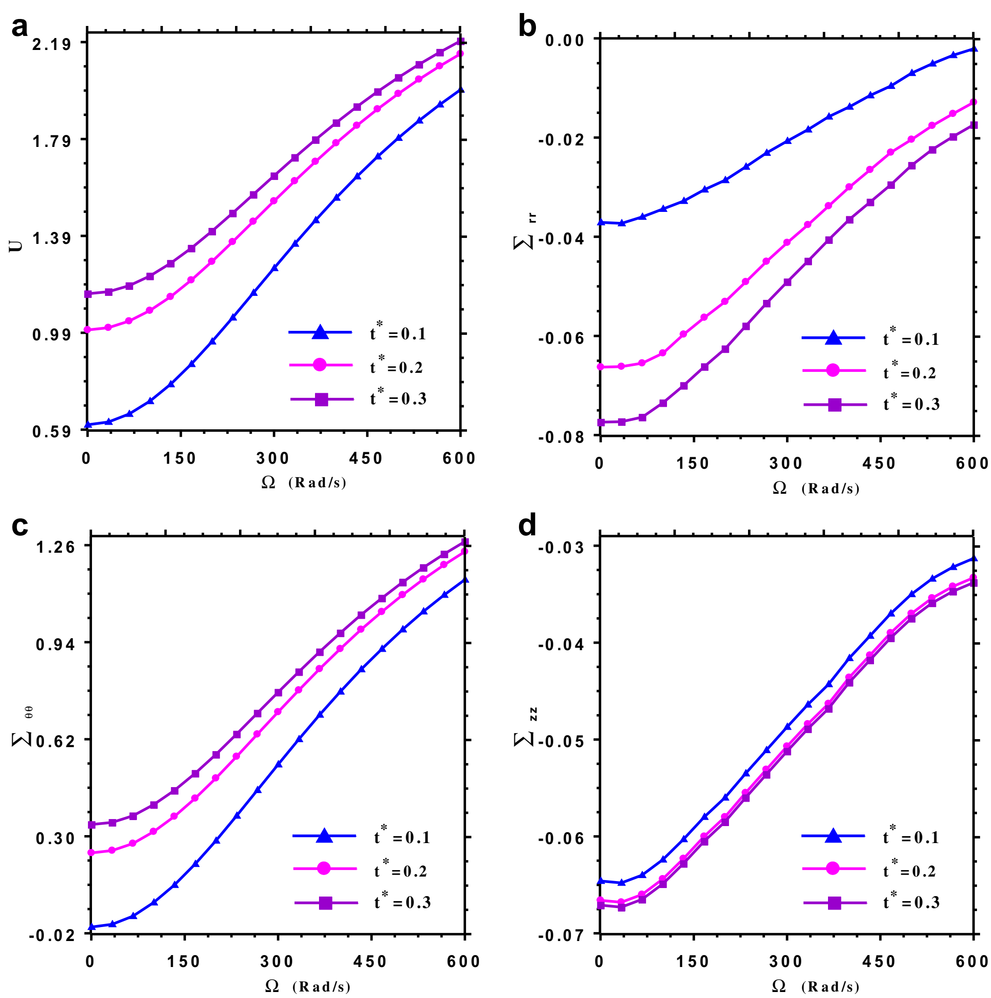

Fig. 10. Comparison of the results for the free ends rotating FG cylindrical shells of model (II) at different time levels $[h/R_o = 0.2, L/R_o = 1, R_o = 1m, N_e = 3, N^e_z = 27, N^e_r = 11, \Omega = 400(\mathrm{Rad/s}), p = 1, \xi = \eta = 0.5]$.

After demonstrating the convergence and accuracy of the present approach, some parametric studies for the rotating FG cylindrical shells subjected to internal dynamic pressure in thermal environment are prepared.

As a first example, comparison between the results of the rotating shell models (I) and (II) with free ends is performed in Fig. 3. The effect of different values of the material graded index (p) on the time history of the results at the point ($\xi = \eta = 0.5$) is shown in this figure. It can be seen that the influence of the material graded index on the non-dimensional parameters of the shell type of model II is much greater than those of model I. Hereafter, otherwise mentioned, the results for the rotating FG shell of model II with free ends are presented.

The effect of the angular velocity on the time history of the results is shown in Fig. 4. It is observable from this figure that increasing the angular velocity of the shell, the non-dimensional radial displacement and tangential stress increase but the non-dimensional radial and axial stresses decrease for all values of the time level.

The influence of temperature dependence of the material properties on the time history of the results is presented in Fig. 5. From the data presented in the sub Fig. 5(a), one can observe that by considering the temperature dependence of the material properties, the radial displacement component increases, which means the reduction of total shell stiffness. In addition, it can be seen from the other subfigures that this effect significantly reduces the radial and axial stress components; however, the reduction of the tangential stress component due to this effect is less than the two former mentioned stress components.

The effects of geometrical parameters (thickness and length) on the time history of the non-dimensional displacement and stress components are presented in Figs. 6 and 7, respectively. It is evident from Fig. 6 that increasing the thickness of the FG shells, the radial displacement and the tangential stress components decrease but the absolute values of the radial and axial stress components increase. From Fig. 7 it can be seen that for the FG shells with a given thickness, the radial displacement and tangential stress components increase by increasing the length of FG shells. But, the radial and axial stress components may increase or decrease by increasing this geometrical parameter; however, they have the same behaviors against the variation of this parameter.

The influence of the different boundary conditions on the time history of the results for the rotating FG cylindrical shell of model (II) is shown in Fig. 8. One can observe that the non-dimensional radial displacement of the FG shells with free and simply supported ends is almost the same but quite different from those of the clamped ends shell. But the non-dimensional radial stress components of the FG shells with clamped and simply supported ends are almost the same.

The through the thickness variations of results at different time level are compared in Fig. 9. It is evident that increasing the time

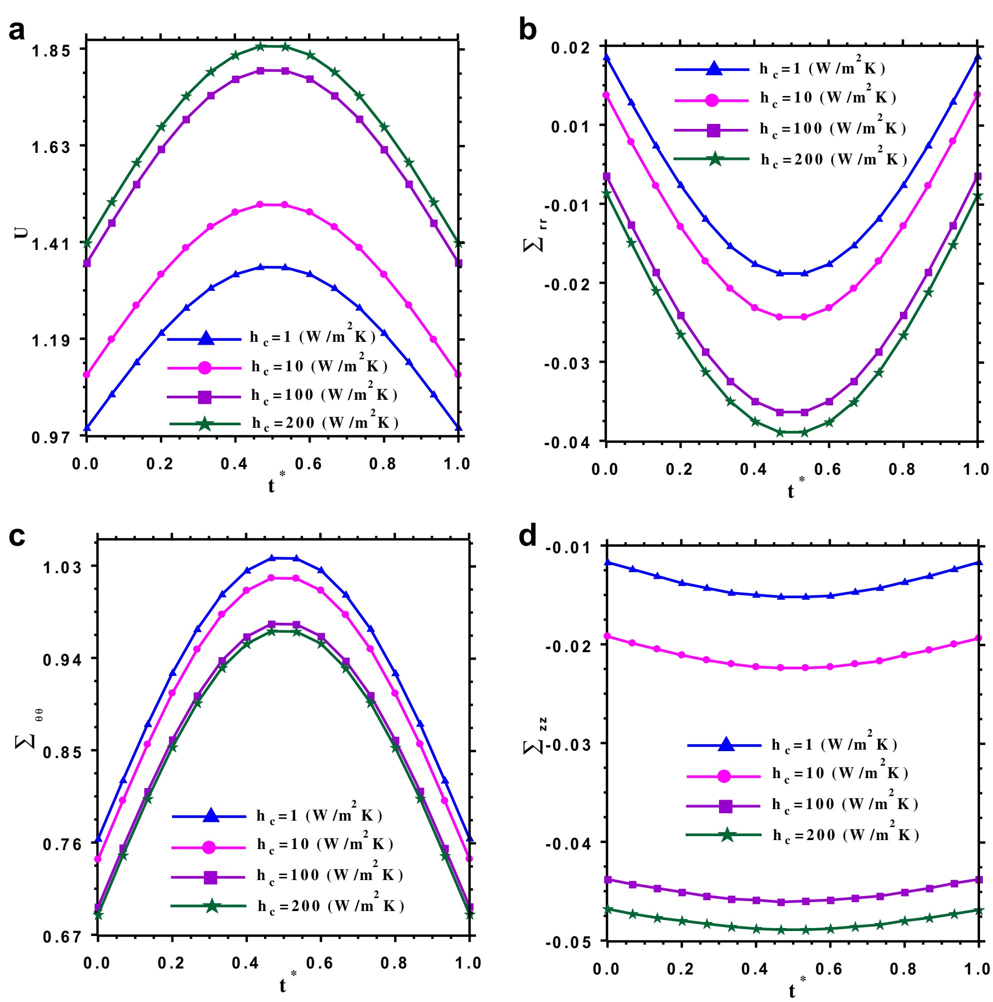

Fig. 11. The influence of convection heat transfer coefficient on the time history of the results for the free ends rotating FG cylindrical shell of model (II)] $[h/R_o = 0.2, L/R_o = 1, R_o = 1m, N_e = 3, N_z^e = 27, N_r^e = 11, \Omega = 400(\mathrm{Rad/s}), p = 1, \xi = \eta = 0.5]$.

level, the non-dimensional tangential stress and displacement components increase but the radial stress component reduces. In addition, it can be seen that at a given point, the axial stress component has almost the same value at different time levels.

The influence of the angular velocity on the results at different time level is shown in Fig. 10. It is observable that the non-dimensional parameters increase monotonically by increasing the angular velocity at all time levels.

As an important thermal parameter, the effect of convective heat transfer coefficient on the time history of the results is exhibited in Fig. 11. It is obvious that this parameter considerably changes the stress and displacement variations in the shell especially for the values of $h_c \leq 100(\mathrm{W/m^2K})$.

4. Conclusion

The transient behavior of the dynamically pressurized rotating laminated FG cylindrical shells in thermal environment were investigated based on elasticity theory. In order to accurately model the variations of the field variables through the thickness of the laminated shell, it was divided into as set of mathematical layers. The material properties were assumed to be temperature-dependent and graded in the thickness direction. The initial thermo-mechanical stresses were obtained accurately by solving the thermoelastic equilibrium equations, which include the inertia forces due to rotation of the shell. After deriving the equations of motion and the related boundary and compatibility conditions at the interface of two adjacent mathematical layers, the DQM in conjunction with Newmark's time integration scheme were employed to discretize them in the spatial and temporal domains, respectively. Fast rate of convergence of the method was demonstrated and its high accuracy with low computational efforts was exhibited by comparing the results with existing solutions in the literature. Then, parametric studies were performed to study the transient behavior of two commonly used laminated FG shells and the results were discussed. From the obtained results, it was concluded that the temperature dependence of material properties, material graded index, the convective heat transfer coefficient, the angular velocity, the boundary condition and the geometrical parameters (length and thickness to outer radius ratios) have significant effects on the displacement and stress components of the rotating laminated FG cylindrical shell. The presented results can be used as benchmark solutions for future works.

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